

1 of 4

- PUBLICATIONS**
1. **Yiran Wang**, Alicia E. Boyd, Lillian Rountree, Yi Ren, Kate Nyhan, Ruchit Nagar, Jackson Higginbottom, Megan L. Ranney, Harsh Parikh, and Bhramar Mukherjee (2026). Ten Core Concepts for Ensuring Data Equity in Public Health. *JAMA Health Forum*. 7(1): e256031. [DOI]
 2. Daniel Y. Dapaah, Shoutaro Arakawa, Gwennyth A. Carroll, **Yiran Wang**, Stewart McLachlin, Mitsuru Saito, and Thomas L. Willett (2025). Advanced Glycation End-product Adducts Alter the Bone Collagen Network and Human Cortical Bone Fracture Resistance. *JBMR Plus*. 10(1): ziaf172. [DOI]
 3. **Yiran Wang**, Martin Lysy, and Audrey Béliveau (2025). Plant-Capture Methods for Estimating Homeless Population Size From Uncertain Plant Captures. *Epidemiology*. 36(3): 319–326. [DOI]
 4. Sean McGrath[†], **Yiran Wang**[†], Yi-Ting Lin, John D. Meeker, Sung Kyun Park, Joshua L. Warren, and Bhramar Mukherjee. A Comparison and Evaluation of Statistical Methods for Mediation Analysis with Mixtures of Environmental Exposures. *BMC Medical Research Methodology*. In Press.
 5. **Yiran Wang**, Martin Lysy, and Audrey Béliveau. Rapid Scaling of Compositional Uncertainty from Sample to Population Levels. *Statistical Methods & Applications*. Under review. [arXiv]
 6. Maxwell Salvatore, **Yiran Wang**, Preeti Syal, Brian Wahl, Harsh Parikh, Vishal Deo, Naveen Kumar Bhatraju, Kaushalya Jayaweera, Nisha Rana, Redoy Ranjan, Fyezah Jehan, Abdullah Yusuf, Karthik Adapa, Mona Duggal, Anurag Agrawal, and Bhramar Mukherjee. The Case for an Integrated Biobanking Initiative in South Asia with Secure and Federated Global Data Access. *The Lancet Regional Health - Southeast Asia*. Submitted.
 7. Daniel Y. Dapaah, Shoutaro Arakawa, Gwennyth A. Carroll, **Yiran Wang**, Stewart McLachlin, Mitsuru Saito, and Thomas L. Willett. Does direct oxidative damage of bone collagen contribute to poor human cortical bone tissue fracture resistance? *Bone*. Submitted.
 8. Lauren Raymond-King, Sean McGrath, Bhramar Mukherjee, **Yiran Wang**, Harsh Parikh, John Rothen, Pamela R. Soulos, Cary P. Gross, and John W. Kunstman. Frailty and Low Utilization of Curative-Intent Surgical Resection in Non-Metastatic Pancreatic Cancer. *Journal of the National Cancer Institute*. Submitted.
 9. **Yiran Wang**, Alicia E. Boyd, Lillian Rountree, Yi Ren, Kate Nyhan, Ruchit Nagar, Jackson Higginbottom, Megan L. Ranney, Harsh Parikh, and Bhramar Mukherjee (2025). Towards Enhancing Data Equity in Public Health Data Science. *arXiv preprint arXiv:2508.20301* [arXiv]
 10. **Yiran Wang**, Yi-Ting Lin, Sean McGrath, John D. Meeker, Sung Kyun Park, Joshua L. Warren, and Bhramar Mukherjee. A Tutorial on Conducting Mediation Analysis with Exposure Mixtures. *arXiv preprint arXiv:2509.10916* [arXiv]
 11. **Yiran Wang**, Martin Lysy, and Audrey Béliveau (2024). Method of Moments Estimation of the Bayesian Credible Interval for Markov Chain Monte Carlo. *Working Paper*.

[†] Co-first authorship

ADDITIONAL PROFESSIONAL EXPERIENCE

Research Assistant

Jan 2020 - Apr 2024

University of Waterloo, Waterloo, ON

- Built a new framework for the plant-capture method to estimate population size by incorporating uncertainty in the plants' assessments and also other factors with Bayesian and frequentist inference approaches.
- Analyzed the stock population, proposed new Bayesian genetic mark-recapture methods, and applied to simulation studies and real-data analysis.
- Advisors: Audrey Béliveau, Ph.D., Martin Lysy, Ph.D.

Curriculum Vitae

Consultant

Sep 2023 - Apr 2024

UW Statistical Consulting & Survey Research Unit, Waterloo, ON

- Advising students and faculty members on how to leverage sound statistical approaches in their analyses, with more active involvement in enhanced service projects.
- Advisor: [Joslin Goh, Ph.D.](#)

TEACHING EXPERIENCE

Teaching Assistant

University of Waterloo, Waterloo, ON

Sep 2019 - Apr 2024

- Held regular office hours, proctored tests, designed and marked assignments and tests for:
 - STAT 202 Introductory Statistics for Scientists Fall 2019
 - STAT 230 Probability Winter 2020, Spring 2020
 - STAT 330 Mathematical Statistics Winter 2021, Fall 2021
 - STAT 331 Applied Linear Models Spring 2020, Fall 2020, Winter 2021
 - STAT 371 Applied Linear Models and Process Improvement for Business Fall 2019, Fall 2020, Winter 2022, Fall 2022
 - STAT 431 Generalized Linear Models and their Applications Spring 2022, Spring 2023
 - STAT 436/836 Introduction to the Analysis of Spatial Data in Health Research Winter 2023, Winter 2024

University of California, Irvine, Irvine, CA

Apr 2018 - Mar 2019

- Held regular office hours, conducted weekly discussion sections, graded homework and examinations for:
 - STATS 7 Basic Statistics Winter 2019
 - STATS 67 Introduction to Probability and Statistics for Computer Science Spring 2018, Fall 2018

CONFERENCE TALKS

Bayesian Mediation Analysis with Latent Exposure Mixtures

2025 Joint Statistical Meetings

Aug 2025

Plant-Capture Methods Using Uncertain Plant Captures

The 38th New England Statistics Symposium

Jun 2025

The Bayesian Young Statisticians Meeting 2025

Apr 2025

The 2025 ENAR Spring Meeting

Mar 2025

The Statistical Society of Canada 2024 Annual Meeting

Jun 2024

2023 Waterloo Student Conference in Statistics, Actuarial Science and Finance

Oct 2023

Genetic Mark-Recapture Methods for Stock Populations

2023 Joint Statistical Meetings

Aug 2023

The Statistical Society of Canada 2023 Annual Meeting

May 2023

EURING Analytical Meeting & Workshop 2023

Apr 2023

The 2022 International Society for Bayesian Analysis World Meeting

Jun 2022

International Statistical Ecology Conference 2022

Jun 2022

The Statistical Society of Canada 2022 Annual Meeting

Jun 2022

The Statistical Society of Canada 2021 Annual Meeting

Jun 2021

EURING Analytical Meeting & Workshop 2021

Jun 2021

Curriculum Vitae

SERVICE

Event and Organization

Instructor & Program Coordinator <i>2025 Big Data Summer Immersion at Yale (BDSY)</i>	Mar 2025 - Jul 2025
Committee Member <i>University of Waterloo Graduate International Student Advisory Committee (ISAC-G)</i>	May 2023 - Apr 2024
Conference Organizing Committee <i>The Eleventh Annual Canadian Statistics Student Conference (CSSC)</i>	Sept 2022 - May 2023
Conference Organizing Volunteer <i>The 2nd Waterloo Student Conference in Statistics, Actuarial Science and Finance</i>	Nov 2021
Student Representative <i>Department of Statistics, University of California Irvine</i>	Oct 2018 - Apr 2019

Mentorship

Mentor <i>University of Waterloo Women in Mathematics Directed Reading Program</i>	Jan 2024 - Apr 2024, May 2023 - Aug 2023
Mentor Volunteer <i>ASA DataFest</i>	Apr 2021, Apr 2022, Apr 2023, Apr 2024

PROFESSIONAL DEVELOPMENT

Intermediate Teaching Workshops: <i>Participation and Equity</i> Yale University	2025
Intermediate Teaching Workshops: <i>Classroom Observation</i> Yale University	2025
Intermediate Teaching Workshops: <i>Teaching International Students</i> Yale University	2025
Advanced Teaching Workshops: <i>Lecturing for Active Engagement</i> Yale University	2025
Intermediate Teaching Workshops: <i>Leading Effective Discussions</i> Yale University	2025
Canadian Statistical Sciences Institute CRT Workshop: <i>Modern Methods in Survey Sampling</i> University of Ottawa	2024
Canadian Ecological Forecasting Initiative Summer School: <i>Forecasting for Decision-Making: An Epidemiological & Ecological Perspective</i> The Fields Institute for Research in Mathematical Sciences	2023
Workshop: <i>NIMBLE virtual short course</i> NIMBLE Development Team	2023
Canadian Statistical Sciences Institute CRT Summer School: <i>Modern Methods in Survey Sampling</i> University of Ottawa	2022